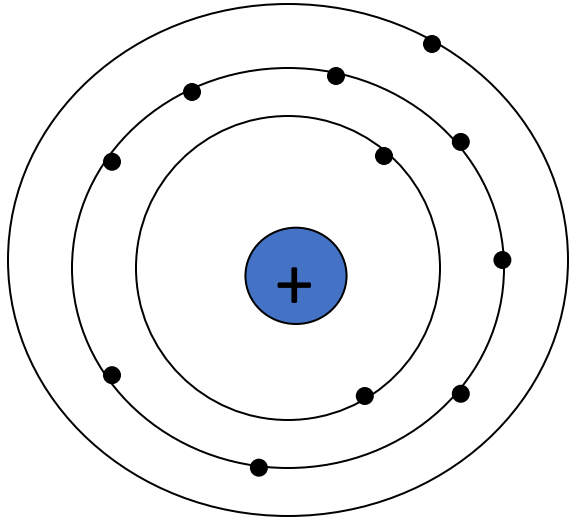
C H A P T E R 7

The Structure of Atoms and Periodic Trends

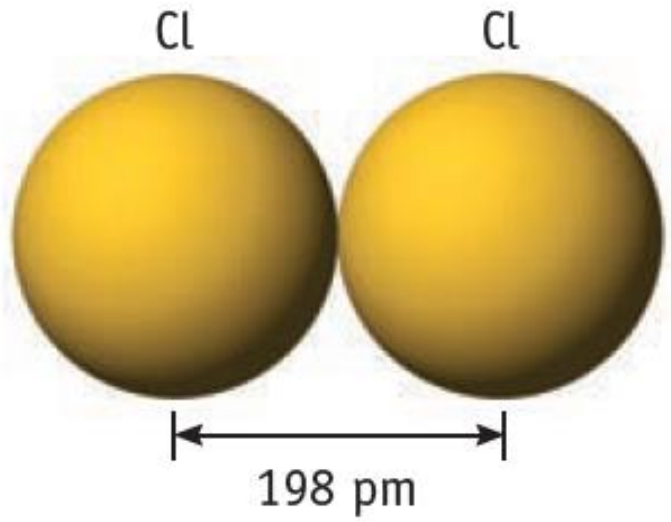
General Periodic Trends

- Atomic and ionic size
- Ionization energy
- Electron affinity



Size - Periodic Trends

- Assume that atoms touch
- Atomic radius is ½ diameter



Effective Nuclear Charge, Z^*

- Z^* is the net charge experienced by a particular electron in a multi-electron atom resulting from a balance of the attractive force of the nucleus and the repulsive forces of other electrons.
- Z^* increases across a period owing to incomplete screening by inner electrons.

Charge felt by 2s electron in:	Li	$Z^* = 3 - 2 = 1$
	Be	$Z^* = 4 - 2 = 2$
	B	$Z^* = 5 - 2 = 3$
		etc…

Effective Nuclear Charge, Z^*

- Screening of valence electrons by core electrons
  - $Z_{eff} = Z - S$
    - Z = atomic number (number of protons)
    - S = number of core electrons (those doing the screening)
  - Na 11 e^- – 10 core e^- = 1+
  - Mg 12 e^- – 10 core e^- = 2+

Atomic Size
For Main Group Elements

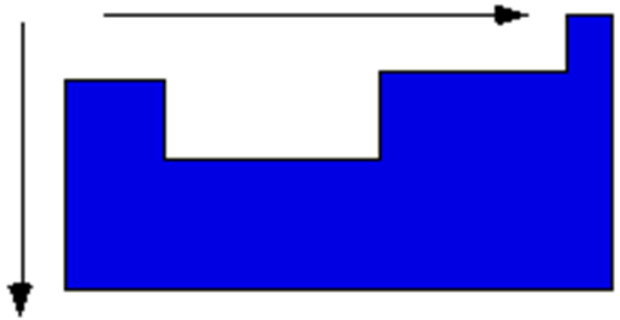
- Atomic Radii generally increase down a group and decrease across a period in the periodic table.

Higher effective nuclear energy

Electrons held more tightly

Larger Orbitals

Electrons held less tightly



Why?

size increases, size and charge increase

Nuclear charge decreases, size increases

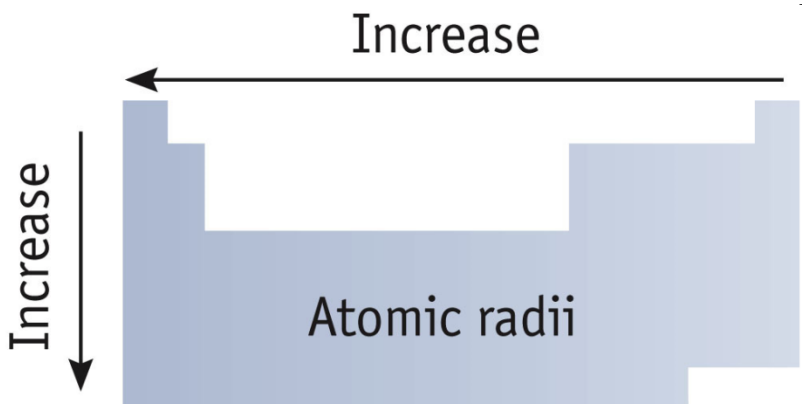
Down a group:

Larger size more important than larger nuclear charge

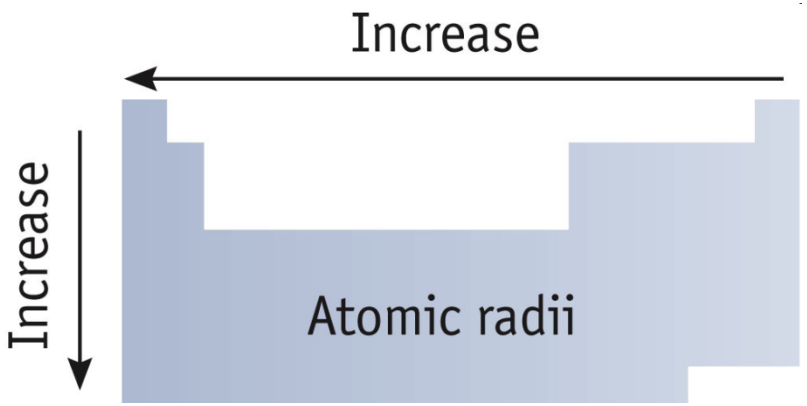
Across a period: n is the same, but the effective nuclear charge increases

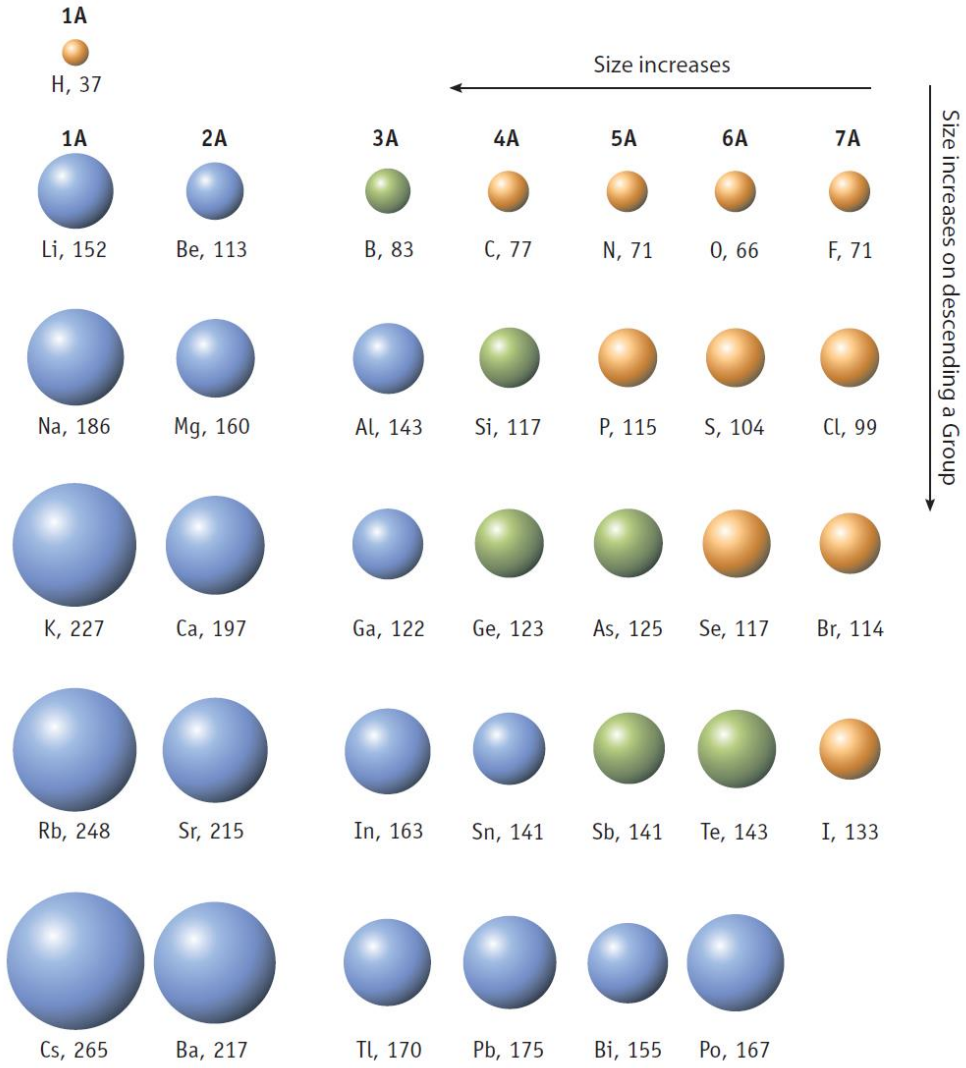
Atomic Size

- **Size goes UP** on going down a group, because electrons are added further from the nucleus, there is less attraction.
- **Size goes DOWN** on going across a period as the effective nuclear charge of the elements increase.



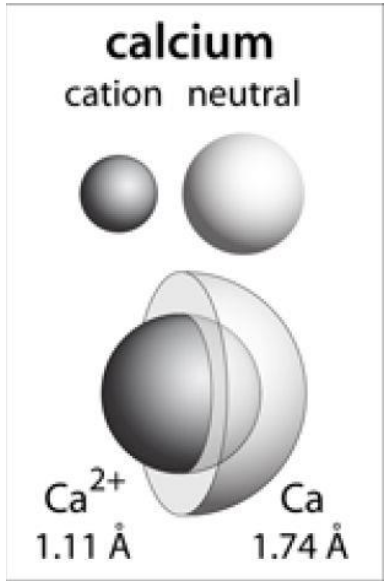
Atomic Radii of Elements

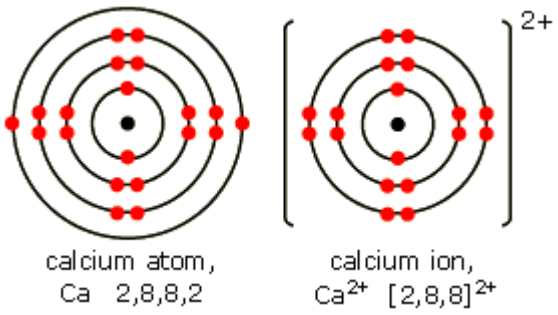


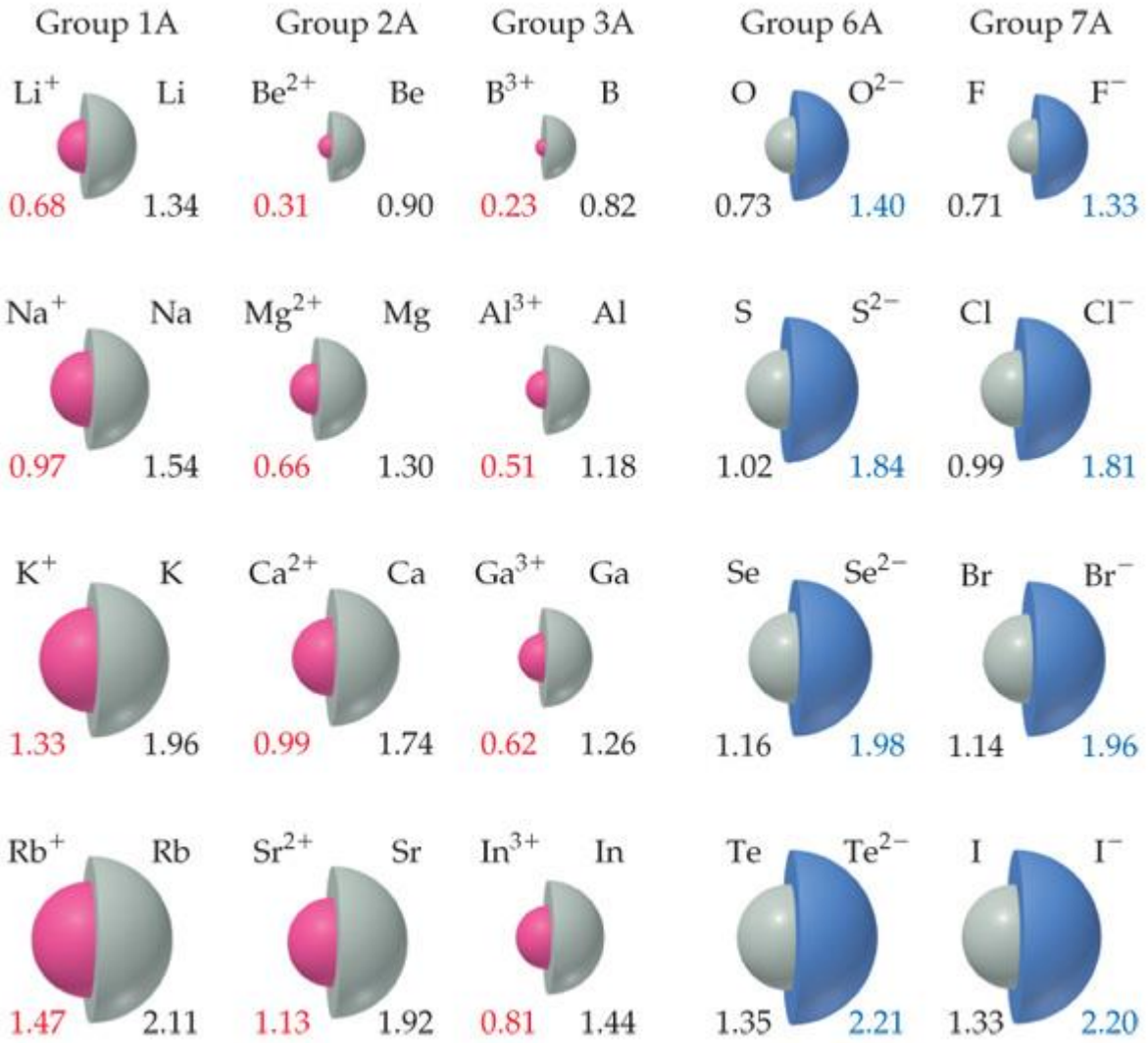


Atomic size of ions

- Cations will be smaller than the respect neutral atom.
- Anions will be bigger than the respect neutral atom.







Problem: Rank the following ions in order of decreasing size?

Na^{+}, N^{3-}, Mg^{2+}, F^{-} O^{2-}

Ion	# of protons	# of electrons	ratio of e/p
Na^{+}	11	10	0.909
N^{3-}	7	10	1.43
Mg^{2+}	12	10	0.833
F^{-}	9	10	1.11
O^{2-}	8	10	1.25

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Since N^{3-} has the highest ratio of electrons to protons, it must have the largest radius.

Since Mg^{2+} has the lowest ratio, it must have the smallest radius.

The rest can be ranked by ratio.

N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+}

Decreasing size →

Notice that they all have 10 electrons: They are **isoelectronic** (same electron configuration) as Ne.

Ionization Energy (*IE*)
$\Delta U \equiv$ ionization energy, *IE*

- The minimum energy required to remove an e^- from the ground state of a gaseous atom
- I_1 - the energy needed to remove the first electron from a gaseous atom.
  - $K(g) + I_1 \rightarrow K^+(g) + e^-$ $\Delta E = +419$ kJ/mol
- I_2 - the energy needed to remove the second electron from a gaseous +1 ion.
  - $K^+(g) + I_2 \rightarrow K^{2+}(g) + e^-$ $\Delta E = +3051$ kJ/mol

Ionization Energy Periodic Trends

decreasing size, **increasing** I

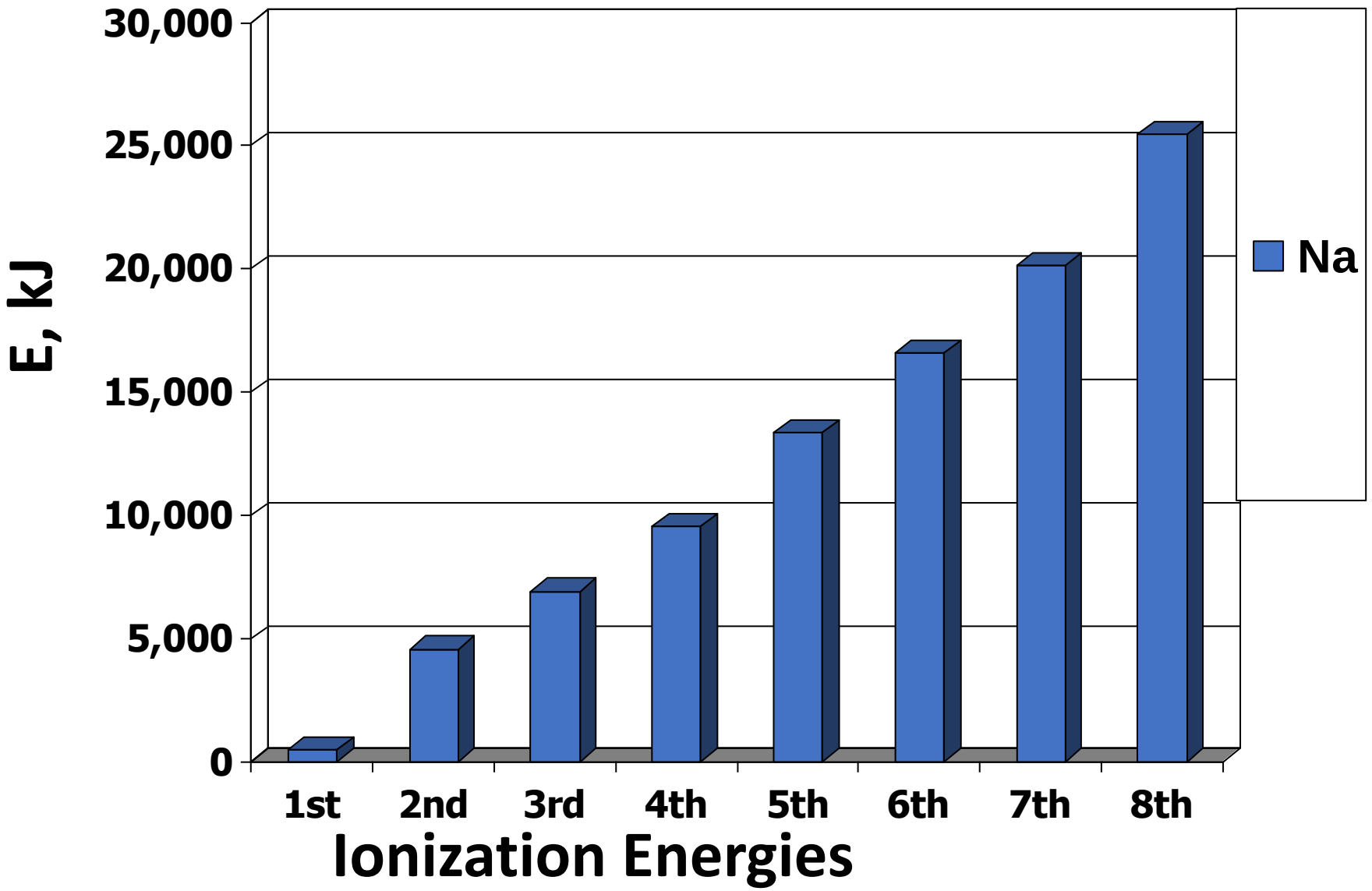
decreasing size, **increasing** I

TABLE 7.5 First, Second, and Third Ionization Energies for the Main Group Elements in Periods 2–4 (kJ/mol)

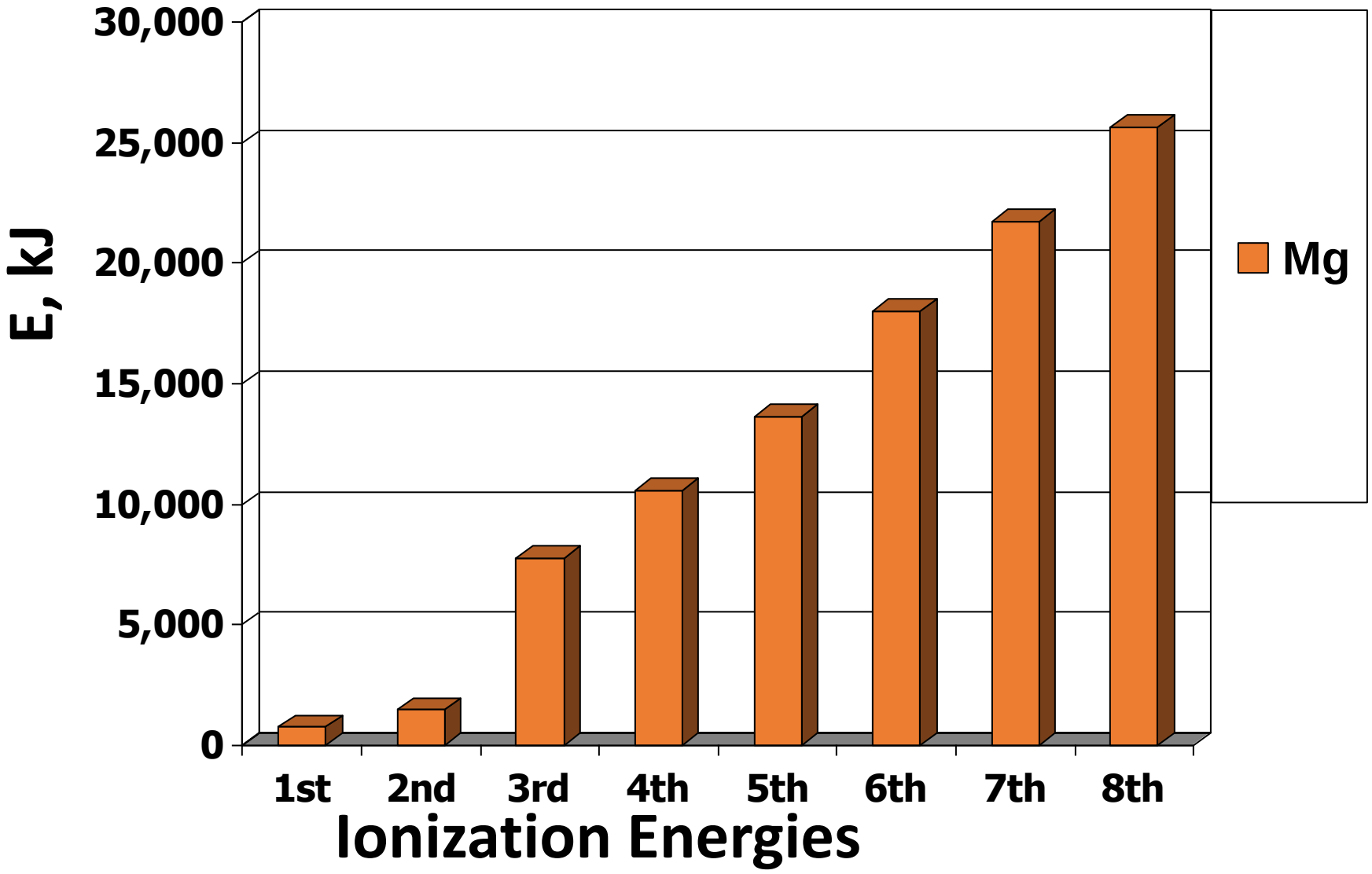
2nd Period	Li	Be	B	C	N	O	F	Ne
1st	513	899	801	1086	1402	1314	1681	2080
2nd	7298	1757	2427	2352	2856	3388	3374	3952
3rd	11815	14848	3660	4620	4578	5300	6050	6122

The general trend: $I_1 < I_2 < I_3 < I_4$...

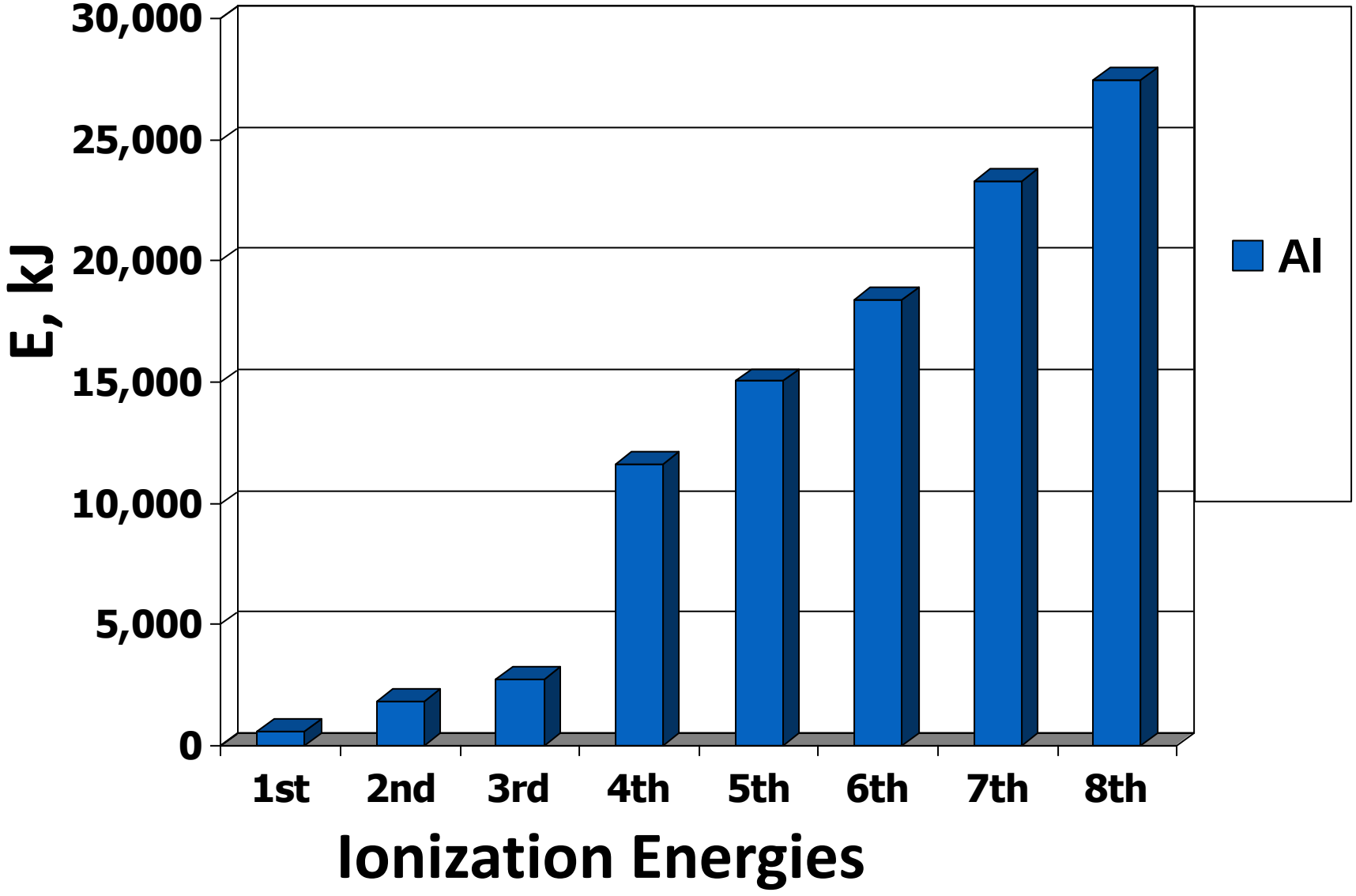
Ionization Energies



Ionization Energies



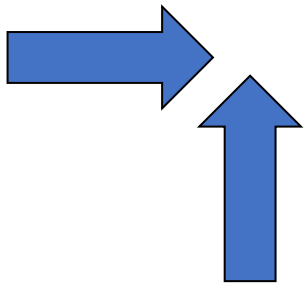
Ionization Energies



Electron Affinity (EA)

- *The energy change for the process of adding an electron to a neutral atom in the gaseous state to form a negative ion*

- $Cl(g) + e^- \rightarrow Cl^-(g)$ $\Delta_{EA}H = -349$ kJ/mol [7.4]
  - remember – sign for energy means the process is favorable
  - TREND Increases as you go



Element	$\Delta_{EA}H$ (kJ/mol)
F	−328
Cl	−349
Br	−324.7
I	−295.2

More on Electron Affinity

- Some elements like the noble gases do not have EA values.
- They have no affinity for electrons.

Ionization Energy vs. Electron Affinity

- I measures ease of losing an e^-
  - always endothermic
- EA measures ease of gaining an e^-
  - often exothermic (but not always)